KEERTHANA M RENJITH

keerthanamrenjith123@gmail.com Keerthana-M-Renjith.github.io linkedin.com/in/keerthana-m-renjith
+91 7592903811

RESEARCH EXPERIENCE

Project Student 2025 – Present
Department of Condensed Matter Physics and Materials Science
Tata Institute of Fundamental Research (TIFR), Mumbai, India
Mentor: Prof. Arnab Bhattacharya

- Experimental study of (001)muscovite mica as a flexible substrate for semiconductor growth using Confocal Raman Spectroscopy and X-ray Diffraction (XRD)
- Growth of Niobium Nitride (NbN) Thin Films on (001)Muscovite Mica by DC Sputtering and Investigation of Strain-Dependent Critical Temperature

ACADEMIC ACHIEVEMENTS AND HONORS

- **Best Research Paper Award**(Oral Presentation) — SPARK'26
  National Level Student Research Conclave — St. Teresa's College, Ernakulam, Kerala
  "Mechanical Bending Effect on the Polarization-Resolved Raman Response of Muscovite Mica." 2026
- **National Initiative on Undergraduate Science (NIUS) – Physics**
  Homi Bhabha Centre for Science Education (HBCSE–TIFR), Mumbai (Selected Participant) 2025

EDUCATION

- **Bachelor of Physics (Honors)**
  Sacred Heart College (Autonomous), Thevara, Ernakulam, Kerala, India 2024 – Present
  Affiliated to Mahatma Gandhi University (MGU), Kottayam, Kerala, India
  CGPA in Physics, Chemistry & Mathematics (4 semesters): **9.64/10**
  CGPA in all courses (4 semesters): **9.31/10**
- **Higher Secondary Education (Science - PCMB)**
  Sacred Heart Higher Secondary School, Thevara, Ernakulam, Kerala, India 2022 – 2024
  Percentage: **99.83%**
- **Secondary Education**
  St.Jude's English Medium Higher Secondary School, Karanakodam, Ernakulam, Kerala, India 2021 – 2022
  Grade: **Full A+**

Medium of Instruction: English (all formal education)

SCHOLARSHIPS

- **Innovation in Science Pursuit for Inspired Research(INSPIRE)** — Department of Science and Technology, Government of India 2024–Present
- **Medha Chhatravriti Yojana Scholarship** — Monetary reward for being among the top 75 girls of Kerala State Education Board in secondary education 2022

SKILLS

- Confocal Raman Spectroscopy: hands-on operation and spectral analysis
- X-ray Diffraction (XRD): peak identification and basic phase analysis
- DC Sputtering: NbN growth
- Origin: curve fitting and Raman tensor analysis
- Python: NumPy, Matplotlib
- VESTA: crystal structure visualization
- LaTeX: academic writing

LEADERSHIP AND ACTIVITIES

- Member — Indian Network of Physics Students (INPS), India 2025 – Present
- Secretary — Sacred Heart Physics Association for Recreation and Kaizen (SPARK) 2025 – 2026